

AI-Generated Image Detection

Based on FerretNet

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- ① Background
- ② FerretNet
- ③ Extension of FerretNet
- ④ References

1 Background

2 FerretNet

3 Extension of FerretNet

4 References

Background of AI-Generated Image Detection



With the development of AI, the images generated by AI are becoming more and more realistic, and people cannot judge the authenticity of the images.

- 1 Background
- 2 **FerretNet**
- 3 Extension of FerretNet
- 4 References

FerretNet: Efficient Synthetic Image Detection via Local Pixel Dependencies

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Abstract

The increasing realism of synthetic images generated by advanced models such as VAEs, GANs, and LDMs poses significant challenges for synthetic image detection. To address this issue, we explore two artifact types introduced during the generation process: (1) latent distribution deviations and (2) decoding-induced smoothing effects, which manifest as inconsistencies in local textures, edges, and color transitions. Leveraging local pixel dependencies (LPD) properties rooted in Markov Random Fields, we reconstruct synthetic images using neighboring pixel information to expose disruptions in texture continuity and edge coherence. Building upon LPD, we propose FerretNet, a lightweight neural network with only 1.1M parameters that delivers efficient and robust synthetic image detection. Extensive experiments demonstrate that FerretNet—trained exclusively on the 4-class ProGAN dataset—achieves an average accuracy of 97.1% on an open-world benchmark comprising 22 generative models. Our code and datasets are publicly available at <https://github.com/xigua7105/FerretNet>

Source: Liang *et al.*, 2025 [1]

Key Contributions

- They proposed **local pixel dependency (LPD) feature map**, which effectively exposes artifacts and anomalies in synthetic images.
- They proposed **FerretNet**, a lightweight network with only 1.1 million parameters, but effectively captures artifacts in LPD feature map.
- They proposed a benchmark dataset called **Syntheal-Pop**, which contains 30K synthetic images from 6 generative models and 30K real images from COCO and LAION-Aesthetics V2 (6.5+).

FerretNet - LPD

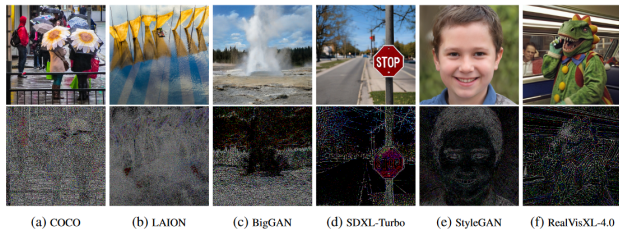


Figure 2: Local pixel dependencies (LPD) comparison between real and synthetic images. Top row: real images (COCO, LAION) and synthetic images (BigGAN, SDXL-Turbo, StyleGAN, RealVisXL-4.0). Bottom row: LPD maps derived from neighborhood-median reconstruction emphasize structural differences.

Insight of LPD

Real images exhibit **stable local pixel statistical consistency**, whereas generative models cannot perfectly reproduce this natural local dependency during latent variable sampling and decoding, thus leaving detectable microscopic damage in local textures and edge continuity.

FerretNet - LPD

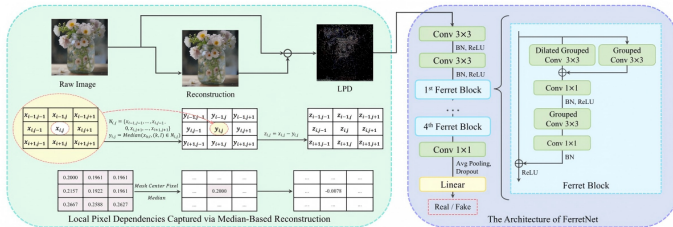


Figure 3: Pipeline of FerretNet: computation of local pixel median discrepancy for artifact representation, followed by lightweight detection using depthwise separable and dilated convolutions.

- $\mathcal{N}_{i,j}$ denotes the set of neighboring pixels located within an $n \times n$ window centered at (i,j) :

$$\mathcal{N}_{i,j} = \left\{ x_{k,l} \mid \begin{array}{l} i-m \leq k \leq i+m, j-m \leq l \leq j+m, \\ (k,l) \neq (i,j) \end{array} \right\}, \quad \text{with } n = 2m+1 \text{ and } m \in \mathbb{Z}^+.$$

- The median-filtered value is computed as:

$$y_{i,j} = \text{Median}(x_{k,l}, (k,l) \in \mathcal{N}_{i,j}), \quad \text{where } \mathcal{N}'_{i,j} = \mathcal{N}_{i,j} \cup \{x_{i,j} = 0\}.$$

- The local pixel dependency (LPD) feature map is defined as: $LPD = I - I'$, where I is original image, I' is median-reconstructed image.

FerretNet - Datasets

Training Dataset

ProGAN 4-class: Only contains four semantic classes (car, cat, chair, horse). Each class contains 18,000 synthetic images generated by ProGAN, paired with an equal number of real images from the LSUN dataset.

Testing Dataset

- ForenSynths [2]: Includes eight representative generative models: ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, Star-GAN, GauGAN, and Deepfake.
- Diffusion-6-cls [3]: Includes six diffusion-based models: DALL-E, Guided, PNDM, VQ-Diffusion, Glide, and LDM.

FerretNet - Question

Methods	Params	w/o LPD	ACC / AP ↑
Xception	20.8M	×	89.8 / 94.1
		✓	95.1 / 98.8
ResNet50	23.5M	×	75.0 / 80.3
		✓	81.1 / 85.6
FerretNet	1.1M	×	86.9 / 93.5
		✓	97.1 / 99.6

Table 1: Ablation study in the original paper

Question

Is this ablation study alone sufficient to prove that LPD and FerretNet indeed captures more artifacts?

- ① Background
- ② FerretNet
- ③ Extension of FerretNet**
- ④ References

Extension of ablation study in FerretNet - Focus on the model

Methods	Params	w/o LPD	ACC / AP ↑
Xception	20.8M	×	89.8 / 94.1
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Table 2: Ablation study in the original paper

Problem

The number of parameters of ResNet50 is 21 times that of FerretNet. Maybe it's just overfitting on the training data.

Extension of ablation study in FerretNet - Focus on the model

Methods	Params	ACC / AP
ResNet18	11.2M	90.5 / 96.0
ResNetLight	1.1M	88.4 / 95.9
FerretNet	1.1M	95.8 / 99.1

Table 3: Results in ForenSynth test set

Methods	Params	ACC / AP
ResNet18	11.2M	86.1 / 94.9
ResNetLight	1.1M	86.7 / 93.1
FerretNet	1.1M	96.3 / 99.0

Table 4: Results in Diffsuion-6-class test set

ResNetLight is a lightweight variant of ResNet18 that keeps only the first two layers (layer1 and layer2) while increasing each layers block count from 2 to 3, removes layer3 and layer4.

Result Analysis

Our supplementary ablation experiments further illustrate that FerretNet is a more powerful network.

Extension of LPD in FerretNet

Our Thinking

The ablation experiment in the original paper does illustrate to some extent that LPD feature map is a better input than the original image, but we believe that only using ablation experiments is not sufficient.

Our Assumptions

If **LPD** feature map indeed exposes more artifacts, then it should have certain statistical features that are easier to capture than on the original image.

Extension of LPD in FerretNet

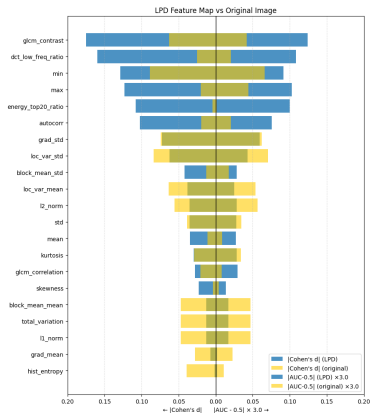
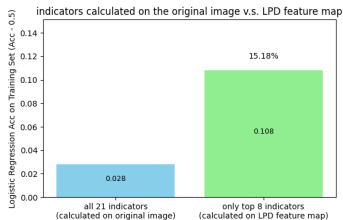


Figure 1: The larger the value, the better the indicator can distinguish between true and fake images.

- 1 We calculated a total of 21 statistical indicators on the original images and LPD feature maps.
- 2 For each indicator, we performed **Cohen's d** and **Mann-Whitney U** tests. The result is shown on the left.
- 3 We identified the top 8 statistical indicators that best differentiate between real and fake images.
- 4 We compared the accuracy of logistic regression using indicators derived from original image and derived from LPD feature map. The result is shown below.



Extension of LPD in FerretNet - Data Selection

Definition of Difficulty

Based on the previous analysis, we define the **Difficulty in Distinguishing** as:

$$D^2(x) = \sqrt{(x - \mu_{real})^T \Sigma_{real}^{-1} (x - \mu_{real})}$$

where

- $x = [x_1, x_2, \dots, x_8]^T$, represents the values of 8 statistical features extracted using LPD feature maps
- $\mu_{real} = \frac{1}{N_{real}} \sum_{i=1}^{N_{real}} x_{real,i}$
- $\Sigma_{real} = \frac{1}{N_{real}-1} \sum_{i=1}^{N_{real}} (\mathbf{x}_{real,i} - \mu_{real})(x_{real,i} - \mu_{real})^T$

D^2 measures the distance of a sample from the center of the real sample.

Extension of LPD in FerretNet - Data Selection

Sampling Strategy

- For real samples, we select the 20% of samples with the largest D^2 to add to our training set.
- For fake samples, we select the 20% of samples with the smallest D^2 to add to our training set.

Extension of LPD in FerretNet - Date Selection Experimental results

Table 5: Comparison of accuracy and average precisions on ForenSynth test set and Deepfake images

Methods	ProGAN	StyleGAN	StyleGAN2	BigGAN	CycleGAN	StarGAN	GauGAN	Deepfake	Mean
Full	100.0/100.0	97.0/99.9	97.4/99.8	93.1/97.5	97.6/99.9	100.0/100.0	92.8/99.7	88.6/96.0	95.8/99.1
Random 20%	99.6/100.0	98.3/99.8	97.2/99.9	83.1/91.8	79.2/92.5	98.6/100.0	65.4/65.9	86.1/93.4	88.4/93.0
Diversity 20%	99.7/100.0	96.2/99.7	96.6/99.9	87.9/95.3	91.8/98.4	98.7/100.0	70.7/78.4	83.2/91.8	90.6/95.4
Difficulty 20%(Ours)	99.4/100.0	94.9/99.9	96.3/99.9	92.8/96.7	96.7/99.6	98.1/99.9	95.6/99.1	89.3/95.9	95.4/98.9

Table 6: Comparison of accuracy and average precisions on Diffusion-6-cls test set

Methods	Dall-E	Glide-100-10	Glide-100-27	Glide-50-27	Guided	LDM-100	LDM-200	LDM-200-CFG	PNDM	VQ-Diffusion	Mean
Full	96.2/99.2	95.2/99.0	92.6/97.8	93.9/97.8	91.1/96.9	98.5/99.7	98.5/99.7	97.8/99.5	99.4/99.9	99.7/100.0	96.3/99.0
Random 20%	74.5/86.6	80.5/93.8	77.9/91.1	79.8/93.3	76.8/88.2	83.8/95.7	83.1/95.1	83.1/95.9	93.3/99.1	99.0/100.0	83.1/93.9
Diversity 20%	68.1/77.2	84.8/94.2	81.9/91.8	84.4/94.1	72.3/85.6	84.5/93.8	84.1/93.1	85.3/94.4	50.5/75.9	99.7/100.0	79.5/90.0
Difficulty 20%(Ours)	88.5/98.0	94.6/98.2	92.6/97.7	94.1/97.8	77.6/92.2	94.3/99.5	92.8/99.2	94.3/99.2	75.6/99.6	100.0/100.0	90.4/98.1

Extension of LPD in FerretNet - Date Selection Experimental results

Result Analysis

- Performance on the in-domain testing set is almost the same as using the full data.
- Performance on the out-of-domain testing set is worse, but still significantly better than random sampling and sampling based on diversity.
- The difficulty indicator we define effectively captures the difficulty of model identification.

Extension of LPD in FerretNet - ContributionS and Weaknesses

Contributions

- We conducted fairer ablation experiments, further demonstrating that FerretNet does better in capturing artifacts.
- We further illustrate the effectiveness of LPD from a statistical perspective.
- Based on LPD, we identify 8 indicators that can effectively distinguish simple and difficult samples.

Weaknesses

- We should try on multiple training sets.
- Our method is more adaptable to in-domain data than out-of-domain data.

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- 3 Extension of FerretNet
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References

- [1] S. Liang, J. Liu, R. Chen, and Q. Guan, “Ferretnet: Efficient synthetic image detection via local pixel dependencies,” *arXiv preprint arXiv:2509.20890*, 2025.
- [2] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, “Cnn-generated images are surprisingly easy to spot... for now,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8695–8704, 2020.
- [3] H. Liu, Z. Tan, C. Tan, Y. Wei, Y. Zhao, and J. Wang, “Forgery-aware adaptive transformer for generalizable synthetic image detection,” 2023.

Thank you for listening !

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